

BRIEF REPORT

Positive Emotional Context Eliminates the Framing Effect in Decision-Making

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Dual-process theories have suggested that emotion plays a key role in the framing effect in decision-making. However, little is known about the potential impact of a specific positive or negative emotional context on this bias. We investigated this question with adult participants using an emotional priming paradigm. First, participants were presented with positive or negative affective pictures (i.e., pleasant vs. unpleasant photographs). Afterward, participants had to perform a financial decision-making task that was unrelated to the pictures previously presented. The results revealed that the presentation framed in terms of gain or loss no longer affected subjects' decision-making following specific exposure to emotionally pleasant pictures. Interestingly, a positive emotional context did not globally influence risk-taking behavior but specifically decreased the risk propensity in the loss frame. This finding confirmed that a positive emotional context can reduce loss aversion, and it strongly reinforced the dual-process view that the framing effect stems from an affective heuristic belonging to intuitive System 1.

Keywords: dual process theories, decision-making, framing effect, emotional context, affective heuristic

Over the last three decades, numerous studies have indicated that individuals often deviate from predictions based on rational choice theory (Cassotti & Moutier, 2010; Evans, 2008; Kahneman, 2003; Reyna & Ellis, 1994). One of the most robust biases in human decision-making is the violation of the principle of description invariance (Tversky & Kahneman, 1981). Invariance requires that changes in the description of the alternative should not alter the preference order. In other words, if a person chooses a medical treatment described as having an 80% survival rate, then logically, that person should still choose the same treatment when it is described as having a 20% mortality rate. However, converging evidence demonstrates that presenting the same option in different formats

alters an individual's decision (Reyna, 2004). Individuals have a tendency to choose differently according to how the suggestion is presented (i.e., corresponding to the well-known framing effect). Typically, when alternatives are phrased in terms of gain, subjects prefer the risk-averse choice. In contrast, when outcomes are phrased in terms of loss, the risk-seeking option is preferred.

To explain this framing bias in decision making, authors have postulated the existence of two distinct systems of reasoning (Evans, 2008; Kahneman, 2003; Kahneman & Frederick, 2007). Dual-process theorists have generally proposed that human decision-making arises from a combination of intuitive/heuristic and analytic/executive processes. In these models, intuitive/heuristic (named "System 1") operations are a form of everyday reasoning. Operations based on System 1 are typically rapid, automatic, parallel, effortless, and often emotionally charged. In contrast, analytic/executive (named "System 2") operations are defined as slow, controlled, serial, effortful, and susceptible to disruption (De Neys, 2006). Kahneman and Frederick (2007) described the framing effect as arising from an affective heuristic belonging to System 1 that operates on the principle that sure gains are particularly attractive, whereas sure losses are particularly aversive (see also Bechara & Damasio, 2005).

In agreement with the affective heuristic hypothesis, both neuroimaging and neuropsychological investigations have shown the involvement of specific brain regions known to be implicated in

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emotional processes, such as the amygdala and the orbitofrontal cortex (see Bechara & Damasio, 2005; Bechara, Damasio, Damasio, & Lee, 1999). Using a monetary gambling task in which participants chose between sure options and risky gambles framed as either a gain or a loss, De Martino, Kumaran, Seymour, & Dolan (2006) examined the neural basis of both the framing effect and the ability to control it. In accordance with the dual-process assumption (i.e., the view that choices are often guided by an initial emotional evaluation belonging to heuristic System 1; see Kahneman & Frederick, 2007), this brain imaging study revealed greater amygdala activation when participants demonstrated a typical framing effect. In psychological terms, this finding suggested that a sure loss triggers a stronger emotional response than a probable loss, leading participants to choose the risk-seeking option. Interestingly, behavioral findings have also shown strong interindividual variability regarding susceptibility to framing, which was not predicted by amygdala activity. Instead, the data revealed a positive correlation between the participant's ability to control framing bias and activation of the orbitofrontal cortex (OFC). Such a result reinforces the view that controlling decision bias depends on inhibiting the initial intuitive, emotional System 1 response. De Martino, Camerer, and Adolph (2009) examined two patients with rare focal bilateral lesions in this region and confirmed that the amygdala plays a key role in generating loss aversion. Moreover, De Martino, Harrison, Knafo, Bird, and Dolan (2008) reported that loss and gain frames induce differential skin conductance responses, which are used as an index of emotional reactivity.

Converging evidence argues in favor of the hypothesis that amygdala activity underlies the susceptibility to framing bias, which corresponds to the operation of an affective heuristic System 1. However, new data support the opposite hypothesis. For example, in contrast to predictions based on neuroimaging investigations, Talmi, Hurlmann, Patin, and Dolan (2010) recently reported that two patients with complete bilateral amygdala degeneration exhibited an intact framing effect. This finding suggested that the amygdala may play a more general role in risk propensity. In the same vein, recent studies have demonstrated that patients with OFC damage could be more rational than neurologically intact participants in economics games or moral judgments (Koenigs et al., 2007; Shiv, Loewenstein, Bechara, Damasio, & Damasio, 2005). In contrast, neuroimaging data showed a positive relationship between rationality and OFC activation (De Martino et al., 2006).

Although the discrepancies in the results of these studies underscore the need for focused research on the role of emotional processes in the framing effect, little is known about the potential impact of a specific emotional context and its positive or negative valence on this decision bias. This lack of knowledge is regrettable because examining the precise influence of different types of emotions on the framing effect may provide experimental arguments in support of the dual-process view. Indeed, if the framing bias arises from an affective heuristic belonging to System 1, the specific emotional context should modulate framing susceptibility.

Despite robust evidence highlighting the fact that a positive mood leads to risk-averse behavior (Blanchette & Richards, 2010; Isen & Geva, 1987; Isen & Patrick, 1983), the specific interaction between framing presentation and a positive emotional context has been neglected. Concerning the impact of a negative emotional

context, Harlé and Sanfey (2007) revealed that sadness induces less acceptance of unfair monetary offers in the ultimatum game compared with a neutral condition. According to the authors' interpretation, subjects with a negative emotional context focused more on the negative consequence of an alternative, whereas a positive emotional context had the opposite effect. Nevertheless, a positive emotional context had no effect on decisions in that game. Finally, recent studies have highlighted that a negative emotional context depends on the specificity of the emotions induced and could lead to opposite risk-taking patterns (Lerner, Gonzalez, Small, & Fishchoff, 2003; Lerner & Keltner, 2001).

The aim of the present study was to uncover the potential impact of both negative and positive subtle emotional contexts on framing bias using an emotional priming paradigm based on the simple presentation of positive or negative affective pictures (i.e., pleasant vs. unpleasant photographs) before and unrelated to the decision task.

Two hypotheses were formulated based on previous research (Blanchette & Richards, 2010; Harlé & Sanfey, 2007). First, if a positive emotional context increases the focus on potential gain, we expected that the affective impact of sure loss (in the loss frame) would be reduced, leading to risk-averse behavior. Therefore, we predicted that the tendency to prefer the risky option in the loss frame should decrease, reducing the framing effect. Second, if a negative emotional context increases the focus on potential loss, then a higher affective impact of the sure loss should be found, resulting in an increased proportion of risky choice in the loss frame. To test these hypotheses, we used a French adaptation of the framing task initially developed by De Martino et al. (2006). An emotional context was induced using standardized pictures from the International Affective Picture System (IAPS; Lang, Bradley, & Cuthbert, 2005), which were presented before each decision.

Method

Participants

A total of 57 participants were recruited from the pool of psychology undergraduate students at Paris Descartes University ($M = 20.8$ years, $SD = 3.21$). Each participant was randomly assigned to one of the three experimental conditions: positive emotional context ($n = 19$), negative emotional context ($n = 19$), and a control group without emotional context ($n = 19$). All participants were naïve regarding the experimental aims.

Design and Procedure

Participants completed computerized decision-making tasks adapted from De Martino et al. (2006). During an instruction period, participants were familiarized with the task and given two practice trials. In each trial, participants received an amount of money in a "monopoly money form" (see Figure 1). Starting amounts were presented for 2500 ms. Subsequently, visual emotional stimuli were presented, each for 5000 ms. A total of 100 photographs selected from the IAPS were included. The IAPS is a collection of standardized photographic materials that include adult ratings (Lang, Bradley, & Cuthbert, 2005). The IAPS pictures were subdivided into two sets according to their valence:

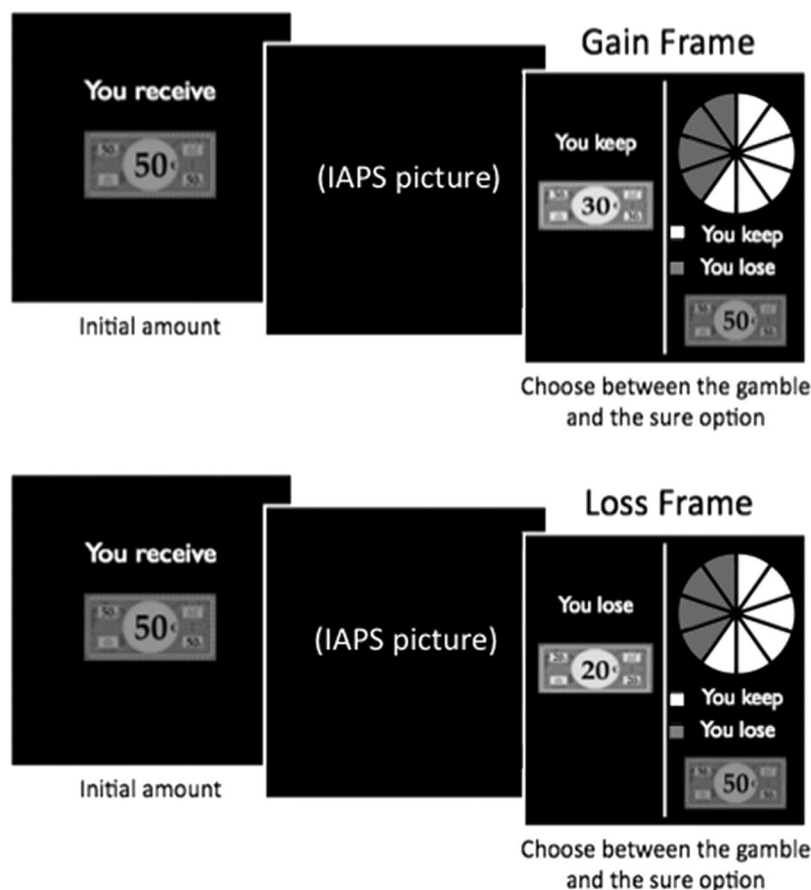


Figure 1. Example of a trial presented in the gain frame and loss frame. In both framing conditions, participants received an initial amount of money in monopoly bill form. An emotional IAPS picture (positive or negative depending on the experimental group) was then presented for 5000 ms. The participants had to choose between a sure option, enabling the retention of a part of the starting amount of money, and a gamble option, enabling the participant to keep or lose the entire initial amount. The sure option was presented in terms of either gain (i.e., “you keep”) or loss (i.e., “you lose”). The expected value of the sure and gamble options was the same, and both frame manipulations were mathematically equivalent.

high (pleasant photography) or low (unpleasant photography).¹ Note that the valence value between the two sets of selected pictures differed significantly (pleasant: 7.73 ± 0.07 , unpleasant: 2.54 ± 0.14 , $t = 72.51$, $p < .0001$). IAPS pictures (1024×768 pixels width and height, respectively) were presented centrally on a 15-in. laptop, and the viewing distance was approximately 60 cm. Participants were then required to choose between a gamble option and a sure option by pressing one of two buttons on a response pad. Each option was framed in terms of either gains or losses. In the gain-framed condition, the sure option enabled the participant to keep part of the starting amount of money, whereas the sure option matched money lost from the starting amount in the loss-framed condition. The gamble option did not change across frame manipulations and was represented with a bicolor “wheel of fortune” depicting the probability of winning or losing the entire starting amount. Expected values were the same for the sure and gamble options, and framing conditions were mathematically equivalent. The procedure for the gain and loss frames was the same except for the words “keep” and “lose” in the sure option.

The initial amount varied between €10 and €50. For each of the two frames, five trials were run, each with five winning probabilities comprising 30%, 40%, 50%, 60%, and 70%, amounting to a total of 50 test trials. The order of the trial presentation was randomized. Probabilities, outcomes, and framing information (gain or loss) were present visually throughout each trial to ensure that all pertinent information was available when participants made their decisions.

¹ The IAPS slide numbers used in the present study were as follows: unpleasant: 9480, 9630, 3301, 2352.2, 2900.1, 9520, 2095, 9252, 2753, 1220, 2800, 1052, 9400, 2455, 9921, 2710, 9584, 6570.1, 1050, 9433, 2710, 9584, 5971, 9570, 3350, 1111, 9220, 9341, 9000, 3005.1, 7380, 3064, 9635.1, 7361, 9530, 2053, 2278, 6313, 2703, 9470, 3015, 9253, 9001, 9040, 1274, 9592, 9910, 3010, 2590, 9410; pleasant: 5836, 2360, 2550, 1610, 2660, 8501, 5831, 8190, 2332, 2150, 1441, 7502, 1460, 1540, 8461, 5833, 2340, 1340, 2224, 1750, 2154, 5760, 2091, 2260, 8420, 2070, 2057, 5621, 7330, 2040, 1500, 8380, 1710, 2370, 5910, 1920, 4641, 8496, 2050, 1440, 2530, 7200, 1590, 2209, 2598, 2071, 2540, 2165, 2080, 8350.

In sum, a trial comprised the following sequence (see Figure 1): In both framing conditions, participants received an initial amount of money in monopoly bill form. An emotional IAPS picture (positive or negative depending on the experimental group) was then presented for 5000 ms. The participants had to choose between a sure option, enabling the retention of a part of the starting amount of money, and a gamble option, enabling the participant to keep or lose the entire initial amount. The sure option was present in terms of either a gain (i.e., “you keep”) or a loss (i.e., “you lose”). Note that the sequence was exactly the same for the control group except that no IAPS pictures were presented.

Results

To examine whether the number of risky choices varied according to the emotional context and framing, we conducted a repeated-measure analysis of variance (ANOVA) with emotional contexts (negative, control, and positive) as a between-subjects factor and frames (gain vs. loss) as a within-subjects factor. This analysis revealed a main effect of frame, $F(1, 54) = 22.91, p < .0001$, indicating that participants made more risky choices in the loss frame condition than in the gain frame condition ($M \pm SD$ gain $41\% \pm 21$; loss $51\% \pm 19$). No main effect of the emotional context was found ($F < 1$). However, a significant interaction was found between emotional context and frame, $F(2, 54) = 6.12, p < .005$, Figure 2. Post hoc comparisons were performed using Tukey's Honestly Significant Difference (HSD) tests, and Cohen's d was used to measure effect sizes.

The Emotional context $\times$ Frame interaction indicated that both the control and negative emotional context groups demonstrated a

standard framing effect. Participants in these two groups chose the gamble option more often in the loss frame presentation (control group: $p < .05, M \pm SD$ gain $37\% \pm 17$, loss $54\% \pm 18, d = .97$; negative group: $p < .0005$, gain 43 ± 23 , loss $56 \pm 22, d = .58$). In contrast, after a positive emotional context, participants did not choose differently between the gain and loss frames ($p < .10; M \pm SD$ gain $42\% \pm 22$; loss $43\% \pm 18$), indicating that these participants were not affected by framing manipulation. Moreover, the participants in the positive emotional context group were more risk-averse in the loss frame compared with the other two groups (planned contrasts, $p < .05$ for the control group, $d = .61; p < .06$ for the negative emotional context, $d = .65$), whereas no significant difference was found in the gain frame.

With regard to response time, a repeated-measure ANOVA with emotional contexts (negative, control, and positive) as a between-subjects factor and frames (gain vs. loss) as a within-subjects factor revealed no main effects of emotional contexts or frames ($F < 1$). There was no significant Emotional context $\times$ Frame interaction. ($F < 1$), suggesting that response time was not affected by framing manipulation and emotional contexts.

Discussion

The aim of the present study was to explore the impact of emotional context on the framing effect during a monetary gambling task. Three major findings emerged from this investigation: (a) both the control (without emotional context) and negative emotional context groups were affected by framing manipulation, (b) the framing effect disappeared following a pleasant emotional context, and (c) positive emotional context did not globally influence risk-taking but specifically decreased risk propensity in the loss frame.

The control group presented a standard framing effect that replicates numerous previous studies (De Martino et al., 2006; Tversky & Kahneman, 1981), indicating that people are often risk-averse when an option is described in terms of gain. Alternatively, people are often risk-seeking when the same options are described in terms of loss. In agreement with neuroimaging investigations (De Martino et al., 2006), the words “keep” and “lose” used in the present study seem to evoke both an emotional response and an automatic tendency to approach or withdraw the sure option (Kahneman & Frederick, 2007; see also Dayan, 2009). This interpretation is also in line with a recent study showing that the incidental presentation of aversive and appetitive conditioned stimuli can modulate the preference for risk in a similar way as framing presentation (Guitart-Masip, Talmi, & Dolan, 2010).

Our results also indicated that a negative emotional context does not affect susceptibility to framing. Indeed, the participants demonstrated a similar pattern of risky choice across framing presentation to the control group following exposure to unpleasant emotional pictures. The lack of an effect of negative emotional context may appear surprising with regard to the emphasis in previous studies on the influence of negative emotions on risk-taking (see Blanchette & Richards, 2010). Specifically, Porcelli and Delgado (2009) recently reported that exposure to acute stress influences financial decision-making by increasing the reflection effect. Moreover, the negative influence of stressor stimuli was observed in decision-making under ambiguity in which information about outcome probabilities was missing and had to be inferred from

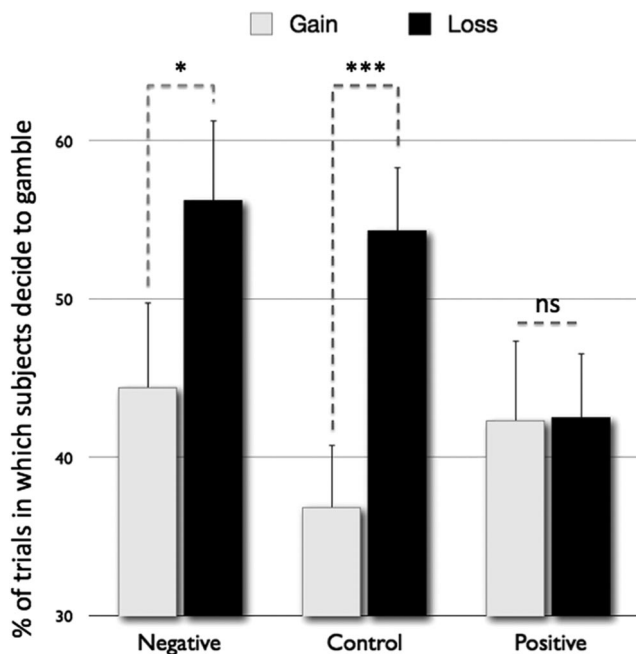


Figure 2. Percentage of test trials for which subjects decided to gamble in the gain and loss frame for the three experimental groups: negative emotional context, control without emotional context, and positive emotional context. * $p < .05$. *** $p < .005$; ns, not significant.

experience. Indeed, using the Iowa Gambling Task (see Bechara, Damasio, Damasio, & Lee, 1999), Preston, Buchanan, Sansfield, and Bechara (2007) demonstrated that the anticipation of giving a public speech reduced the participants' ability to shift toward advantageous decision-making. Nevertheless, the IAPS pictures used in the present study were chosen based on their valences (positive or negative) rather than specific emotions that they could also trigger (such as disgust, anger, or fear). These specific negative emotions were studied by Lerner and Keltner (2001), who observed opposite effects of anger and fear on risk judgments. Although fear increased risk estimates, anger led to the opposite effect. In the same vein, the choices of fearful people are often known to be risk-averse, whereas the choices of angry people are often risk-seeking (Lerner et al., 2003). Taken together, these findings suggest that additional studies will be necessary to determine the impact of specific negative emotions on the framing effect.

The major result of the present study was that framing susceptibility disappeared after the induction of a positive emotional context. In other words, the presentation format in terms of gain or loss no longer affected subjects' decision-making following exposure to emotionally pleasant pictures. Interestingly, a positive emotional context did not globally influence risk-taking behavior but specifically decreased risk propensity in the loss frame. This finding confirmed that a positive emotional context can reduce loss aversion and strongly reinforced the dual-process view that the framing effect arises from an affective heuristic belonging to intuitive System 1.

Using intentional reinterpretative regulation strategy, Sokol-Hessner et al. (2009) reported a comparable reduction of individuals' loss aversion. Interestingly, the authors demonstrated that cognitive regulation techniques (such as "thinking like a trader") may reduce arousal to loss, leading to a diminution of the spontaneous tendency to avoid choices that can lead to losses. Further neuroimaging investigations focusing on the influence of the positive emotional context or regulation strategy will be necessary to provide crucial data about amygdala and orbitofrontal activations during decision-making. In particular, a functional magnetic resonance imaging (fMRI) study is necessary to determine whether the positive emotional context before the choice reduces amygdala reactivity to the framing manipulation or increases activity in the orbitofrontal cortex, leading to enhanced control of framing bias.

In conclusion, our findings demonstrate that a task-irrelevant subtly positive emotional context can have a significant impact on decision-making. Specifically, positive emotional pictures presented before an individual makes a choice are sufficient to eliminate the framing effect by reducing the aversive aspect of sure loss. As such, the present study provides new fuel for the current debates on the dual-process view that the framing effect arises from an initial emotional evaluation.

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